

Youssef Ahmed

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TECHNICAL SKILLS

Languages & Frameworks: Java, Python, C, C++
Libraries & Tools: JUnit 5, Java Swing, UML, Git
Cloud & Infrastructure: AWS, Terraform (Infrastructure as Code), Docker, Linux

PROJECT BASED WORK EXPERIENCE

- Ontology Researcher & Information Architect, Roots Solutions & Automation** May – Aug 2026
- Designed a universal ontology as a detailed context layer for AI, modelling an organization's people, assets, documents, operations, events, commitments, and transactions.
 - Developed a framework enabling AI to understand organizational structure, relationships, and processes across different companies and industries
- AI Analytics Tool Developer, Seen Education | Python** Aug 2026 – Present
- Developed a Python based platform to analyze and benchmark student performance across grades, schools, and national datasets.
 - Built automated scale score calculations, statistical comparisons, and performance benchmarking.
 - Automated the creation of client-ready visual reports for schools, including charts, comparisons, and executive summaries.
- Development Operations, Seen Education | AWS, Terraform** Aug 2026 – Present
- Provisioning AWS infrastructure using Terraform (Infrastructure as Code)
 - Designed the infrastructure to support deployment across schools throughout the Middle East, serving thousands of students.

PROJECTS

- Valet Parking App (Academic), Software Construction | Java, JUNIT** Jan – Apr 2026
- Developed a Java desktop application for valet parking management featuring a Java Swing GUI and console UI, both backed by the same model, with functionality for vehicle check-in, slot allocation, retrieval, and real-time wait time estimation.
 - Achieved high unit test coverage using JUnit 5 across model and persistence layers, and documented system architecture through a UML class diagram capturing all associations, multiplicities, and interface relationships.
- HR Screening Automation (personal) | n8n, Docker, API Automation** May – May 2026
- Built an automated HR workflow in n8n that monitors incoming emails for resumes and processes candidate applications against specified selection criteria.
 - Connected multiple workflow steps to transform incoming email data into structured candidate information and automate the screening process
- Parking Lot Management Database Automation (personal) | n8n, Supabase, Twilio, Docker, API Automation** Jun – Jun 2026
- Developed an n8n workflow to process parking activity and automatically update vehicle records as cars enter or leave.
 - Integrated workflow automation with Supabase to keep operational data synchronized automatically and used Twilio to notify the user with updates.
 - Self-hosted the n8n runtime in Docker, managing container configuration, environment variables, and persistent volumes so workflows and credentials survived restarts.

TECHNICAL EXTRACURRICULAR ACTIVITIES

- Internship, Roots Solutions & Automation** Jun 2022 – Jul 2024
- Researched multi-view 3D reconstruction methods for a proposed multi-camera rig that scans parts as they move through the capture volume.
 - Visited industrial sites across Egypt and the UAE to document operational issues and the automation solutions each company planned to adopt.
- Amazon Robotics Hackathon | UBC, Vancouver | Sep 2026** Sep 2026
- Placed 14th of 49 teams by implementing and benchmarking multiple optimization algorithms in Python to minimize robot step count and runtime on a real-world problem set by Amazon Robotics engineers.

EDUCATION

2nd Year Bachelor of Science, Computer Science, Statistics Combined Major (average: 88.5 %) Sep 2025 – May 2030 (Expected)
University of British Columbia, Vancouver, BC